

FIELD THEORY PART-II

Phase 9: Poynting, Polarization and Concepts

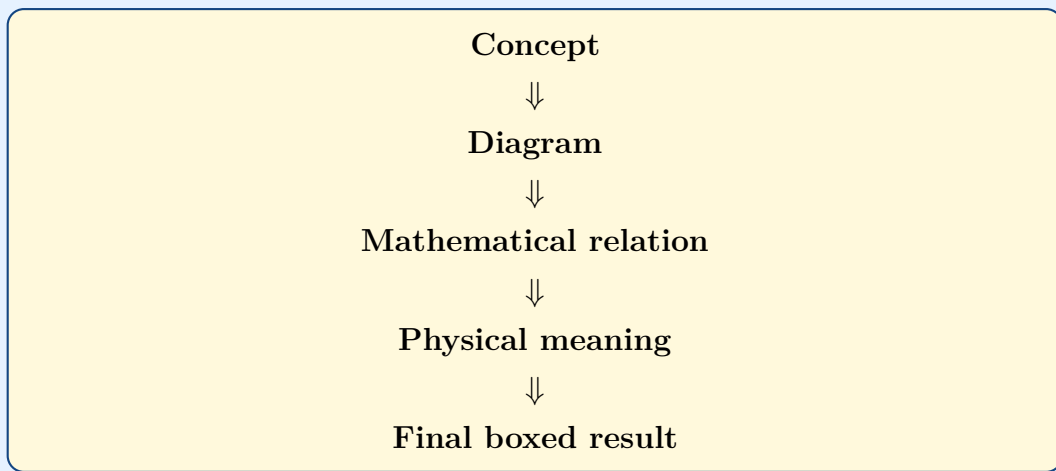
Energy flow, polarization, skin effect, antenna and waveguide

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



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1 Aim of Phase 9

Main Goal

This phase covers Poynting vector, Poynting theorem, polarization, skin effect, antenna and waveguide concepts.

2 Poynting Vector

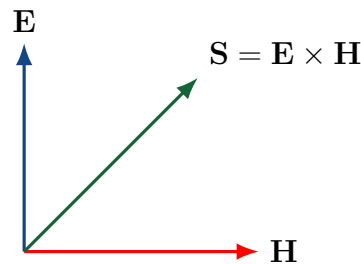
Definition

Poynting vector represents electromagnetic power flow per unit area:

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Unit:

$$W/m^2$$



Direction of EM energy flow

Figure 1: Poynting vector direction.

3 Poynting Theorem

Energy Density

Electromagnetic energy density is:

$$u_{em} = \frac{1}{2}\epsilon E^2 + \frac{1}{2}\mu H^2$$

Differential form of Poynting theorem:

$$\nabla \cdot \mathbf{S} = -\frac{\partial u_{em}}{\partial t} - \mathbf{E} \cdot \mathbf{J}$$

Integral form:

$$\oint_S \mathbf{S} \cdot d\mathbf{S} = -\frac{\partial}{\partial t} \int_V u_{em} dV - \int_V \mathbf{E} \cdot \mathbf{J} dV$$

4 Meaning of Poynting Theorem

Physical Interpretation

Power leaving a volume is equal to rate of decrease of stored electromagnetic energy minus ohmic loss.

5 Polarization

Definition

Polarization describes the time variation of the electric field vector of an electromagnetic wave at a fixed point.

Path traced by the tip of electric field vector

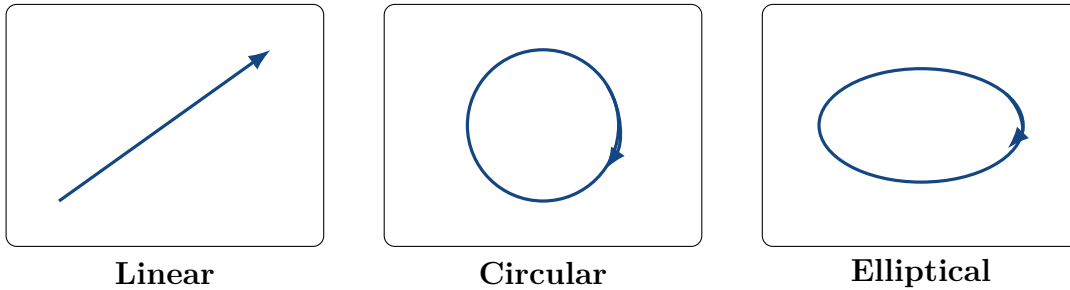


Figure 2: Linear, circular and elliptical polarizations.

6 Types of Polarization

Linear Polarization

If the electric field vector always remains along a fixed straight line, the wave is linearly polarized:

$$\mathbf{E} = E_0 \cos(\omega t - \beta z) \hat{\mathbf{x}}$$

Circular Polarization

If two perpendicular components have equal amplitude and phase difference of 90° , the tip of $\mathbf{E}$ traces a circle:

$$E_{0x} = E_{0y}, \quad \delta = \pm \frac{\pi}{2}$$

Elliptical Polarization

If amplitudes are unequal or phase difference is general, the tip of $\mathbf{E}$ traces an ellipse. Linear and circular polarization are special cases of elliptical polarization.

7 Skin Effect

Meaning

Skin effect means alternating current tends to concentrate near the surface of a conductor at high frequency.

8 Antenna Concept

Meaning

An antenna converts electrical signals into electromagnetic waves and receives electromagnetic waves as electrical signals.

9 Waveguide Concept

Meaning

A waveguide guides electromagnetic waves from one point to another with small radiation loss.

10 Exam Writing Template

For Short Notes

Use:

1. Definition.
2. Formula.
3. Unit.
4. Physical meaning.
5. Simple diagram.

11 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.